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GENESIS

Genesis Ideathon: Final Presentation

Lease-Based Emergency Dispatch Failover

Safe Ambulance Assignment When Control-Room Connectivity Fails

Team ID: *GEN26-A0H-I141*

Team Name: *SHINCHAN*

Domain: *Disaster Management & Resilient Emergency Communication*

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Presentation Outline

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Problem Statement

During floods, cyclones and building collapses, cellular towers, internet links and power can be **damaged or congested**. Ambulances, hospitals and rescue teams then lose the updates they depend on: **where victims are, how urgent each case is, and who is already responding**.

Central question: How can emergency responders exchange essential, prioritised information – and know that it was received – when conventional communication networks fail?

Problem Understanding

- **Who experiences this problem?**
  - People awaiting rescue after a flood, cyclone or collapse
  - Ambulance crews, hospital emergency desks, fire and rescue teams
  - Incident coordinators managing many requests at once
- **What causes the problem?**
  - Damaged towers, cut fibre and power outages
  - Network congestion when everyone calls at once
  - Tools that only work through a central server or the internet
- **How does it affect users or stakeholders?**
  - Requests, locations, urgency and status fragment across teams and channels
  - The same victim is reported twice, or a request is never received
  - Nobody can tell whether a message actually reached its recipient

Existing Gaps & Challenges

- **Gap 1 – Infrastructure dependence:** many coordination apps and cloud dashboards stop working when internet access or the central server is unreachable.
- **Gap 2 – Unstructured information:** what does keep working (voice radio, chat apps, off-grid messengers) carries free text or voice, not structured, prioritised requests with a delivery state.
- **Main unresolved challenge:** a phone-based way to create, prioritise, store and deliver emergency requests offline, *with confirmation that the receiving app saved them* and without duplicate incidents.

What already exists: dedicated radio networks are built for disasters; Meshtastic (LoRa radios) and Briar (Bluetooth/Wi-Fi) move messages without the internet. Our system does not replace these; it targets the structured-request gap above.

Design principle: Local communication first. Internet connectivity when available.

Our Proposed Solution

Emergency Dispatch Failover: a local-first Android app for responders.

How it works

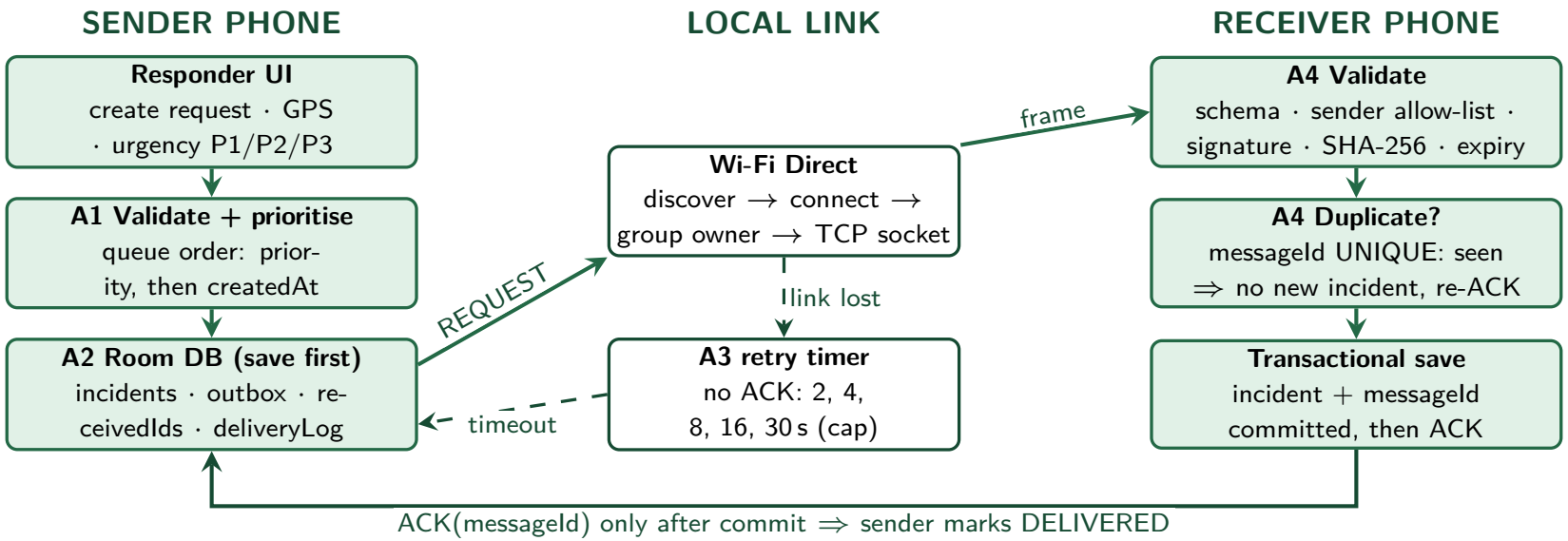
1. Responder creates a request: GPS location, urgency (P1–P3), essential details.
2. The app validates it, ranks it and **saves it on the phone first**.
3. When a peer phone is in range, it connects over **Wi-Fi Direct** and sends.
4. The receiver checks and saves it, then returns an **ACK**.
5. No ACK $\Rightarrow$ the message stays queued and is retried.

Phase 1: direct phone-to-phone.
Phase 2: bounded relay, optional sync.
Limits: no link $\Rightarrow$ no delivery; GPS can be stale.
Complements, never replaces, emergency services.

Algorithm	Input $\rightarrow$ output	Used by
A1 Rule-based prioritisation	request + chosen urgency $\rightarrow$ next message to send	priority queue
A2 Persistent store-and-forward	new message $\rightarrow$ saved outbox entry, sent when a peer is reachable	Room DB + outbox
A3 ACK + exponential backoff	sent message, ACK or timeout $\rightarrow$ DELIVERED, retry or EXPIRED	delivery manager
A4 UUID dedup + SHA-256 check	received frame $\rightarrow$ saved once + ACK, or rejected	receiver validator
A5 Bounded store-carry-forward *(Phase 2)*	message + relay peer $\rightarrow$ forwarded copy within TTL, hop and copy limits	relay module

Privacy by design (planned): only essential patient fields; only authorised responders can open records; database encrypted on the phone.

Technical Approach



Outbox state: QUEUED → SENT → DELIVERED | no ACK → QUEUED (retry) | expired → EXPIRED

Phase 2 (planned) – A5 bounded store-carry-forward: Device A → relay phone → Device B → recipient; limits: expiresAt, hopCount ≤ 3, ≤ 2 copies, seen IDs dropped. Optional sync with an authorised server when internet returns. Wi-Fi Direct alone forms one group, not a mesh.

REQUEST frame: messageId (UUIDv4), incidentId, senderId, messageType (REQUEST/UPDATE/CANCEL/ACK), createdAt, expiresAt, priority, location, payload, hopCount, checksum = SHA-256(canonical fields), signature (ECDSA P-256). deliveryStatus is local only.

Tech Stack (planned) **Frontend:** Android, Kotlin **Database:** Room/SQLite, SQLCipher at rest **Backend:** on-device foreground service; sync server only in Phase 2 **Network:** Wi-Fi P2P API + TCP sockets **Security:** Android Keystore keys, signed frames, encrypted channel **Location:** device GPS

Innovation & Uniqueness

What is new or different? Not a new radio or a new chat app: one responder workflow that combines structured, prioritised requests (A1), save-first queuing (A2), ACK-confirmed delivery (A3) and duplicate-free, integrity-checked storage (A4) on ordinary phones.

Option	Main purpose	Extra hardware	No internet?	Prioritised incidents
Voice radio	voice coordination	radio sets	yes	manual
Meshtastic	text over LoRa mesh	LoRa radios	yes	not built in
Briar	secure messaging	none	yes (Bluetooth / Wi-Fi)	not built in
Cloud dispatch apps	dashboards, tracking	none	no	yes
Our system	**structured emergency requests**	**none in Phase 1**	**yes (direct link)**	**yes, with ACK + dedup**

Why it suits the problem: it keeps working with only the phones responders already carry, and every request has a visible state (queued, delivered, expired).

Honest claim: the building blocks (Wi-Fi Direct, store-and-forward, ACKs, UUIDs, SHA-256) are established; the contribution is their integration for responders. Comparison is based on public product descriptions; novelty still needs a wider literature check.

Feasibility & Scalability

Feasibility – MVP build order (phones only)

1. Request form, GPS capture, A1 priority queue
2. Room schema + outbox; save before send (A2)
3. Wi-Fi Direct discovery, connect, socket exchange
4. Frame codec, SHA-256, signatures, A4 receiver
5. ACK, A3 backoff, expiry, delivery log
6. Restart / out-of-range tests on 4 phones

Scalability – Phase 2 roadmap

- A5 relay on spare Android phones first
- Custom relay boxes only after a compatible link is proven (ESP32 has no documented Wi-Fi Direct; would need softAP or BLE)
- Authorised hospital/server sync, then wider pilots

```
sender loop
next = outbox.where(state=QUEUED,
  nextTry<=now).orderBy(priority,createdAt)
send(next); next.state = SENT
on ACK(id):  state = DELIVERED
on timeout:  k++; nextTry = now+min(2^k,30)s
  state = QUEUED
if now > expiresAt:  state = EXPIRED

receiver
if !valid(schema,sender,sig,sha256)
  or expired:  reject+log
tx{ if exists(messageId):  duplicate
  else save(incident, messageId) }
send ACK(messageId)   // only after commit
```

Risks → mitigation

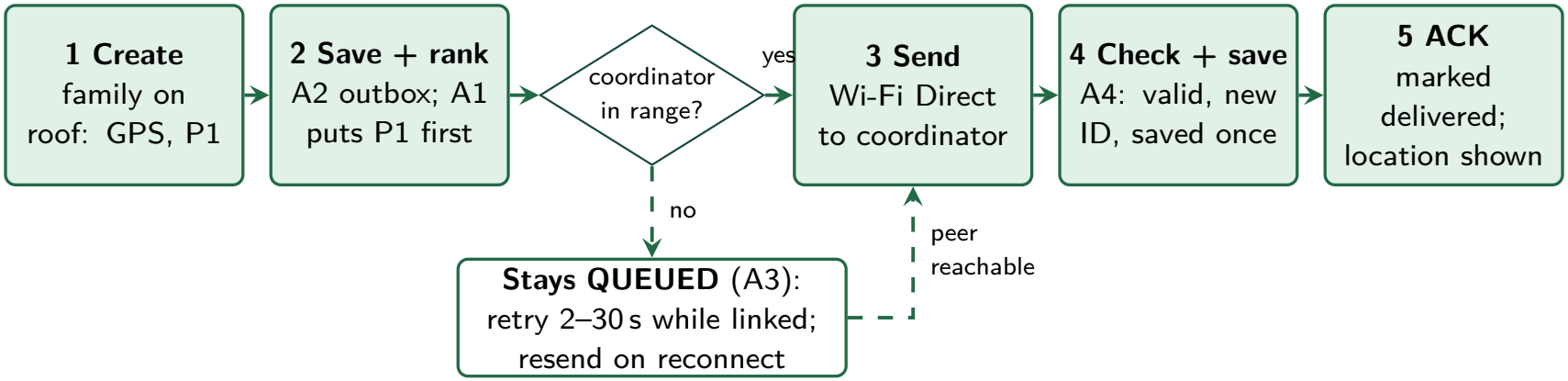
- Range / walls → measured range test first
- Connection prompts, OEM differences → test 4 models
- Lost ACK → idempotent re-ACK via messageId
- Patient data → minimal fields, encryption, dummy test data

Cost for Product

- Development or hardware cost:
  - **Phase 1 MVP:** Rs. 0 software (Android Studio, Kotlin, Room, open-source libraries); 4 existing Android phones (not costed)
  - Field-test extras (power banks, cables, tape measure): Rs. 0–1,500
  - **Phase 2 (only if needed):** 3 spare Android phones as relays (reuse; not costed), or 3 ESP32 boards (listed from about Rs. 345 each, excl. tax) + batteries and enclosures: about Rs. 2,500–5,000
- Hosting or maintenance cost: Rs. 0 in Phase 1 (no server). A Phase 2 sync server would add a recurring cost, to be estimated when designed.
- Estimated total: **Phase 1: Rs. 0–1,500** | Phase 2 add-on: about Rs. 2,500–5,000

Revised from the earlier Rs. 3,300–8,000, which budgeted relay hardware the MVP does not need. Estimates, not quotations; ESP32 price from an IndiaMART listing (Oct 2026). Excludes phones and team labour.

Use Case 1: Flood or Cyclone Evacuation



Situation: river flooding cuts mobile coverage in a village; boats and field teams need to know which homes still have people.

Actors: field responder, incident coordinator; affected residents are the people being reported.

Constraint: no internet; the coordinator is only sometimes within Wi-Fi Direct range.

Algorithms: A1, A2, A3, A4.

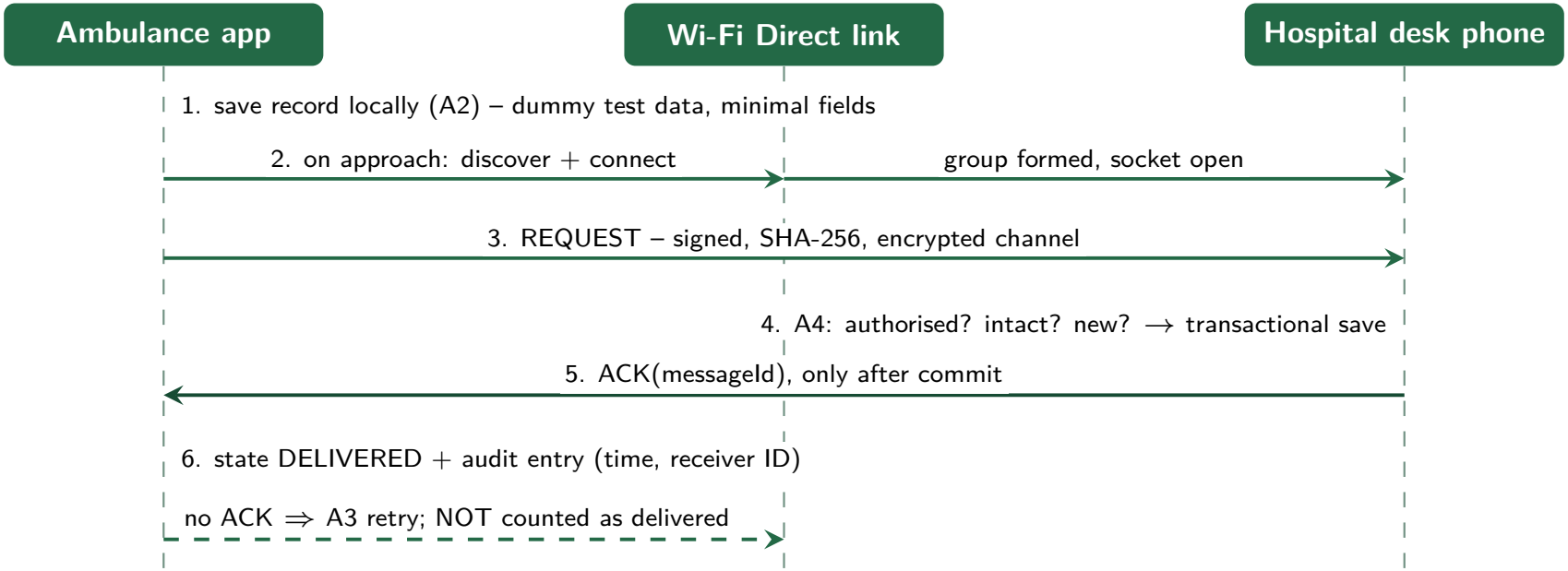
If the link fails: the request is never lost; it waits in the outbox until a peer is reachable or it expires.

When it returns: queued P1 requests go first; repeats are de-duplicated by messageId.

Outcome: the coordinator sees each pending evacuation, its priority and last reported location.

Limits: GPS may be stale or inaccurate; no delivery while phones stay apart.

Use Case 2: Ambulance-to-Hospital Coordination



Situation: an ambulance carries a critical patient during a regional outage; the hospital needs details before arrival.

Actors: ambulance crew, hospital emergency desk.

Constraint: a direct link exists only once the ambulance is within range of the hospital phone.

Algorithms: A1 (P1 first), A2, A3, A4.

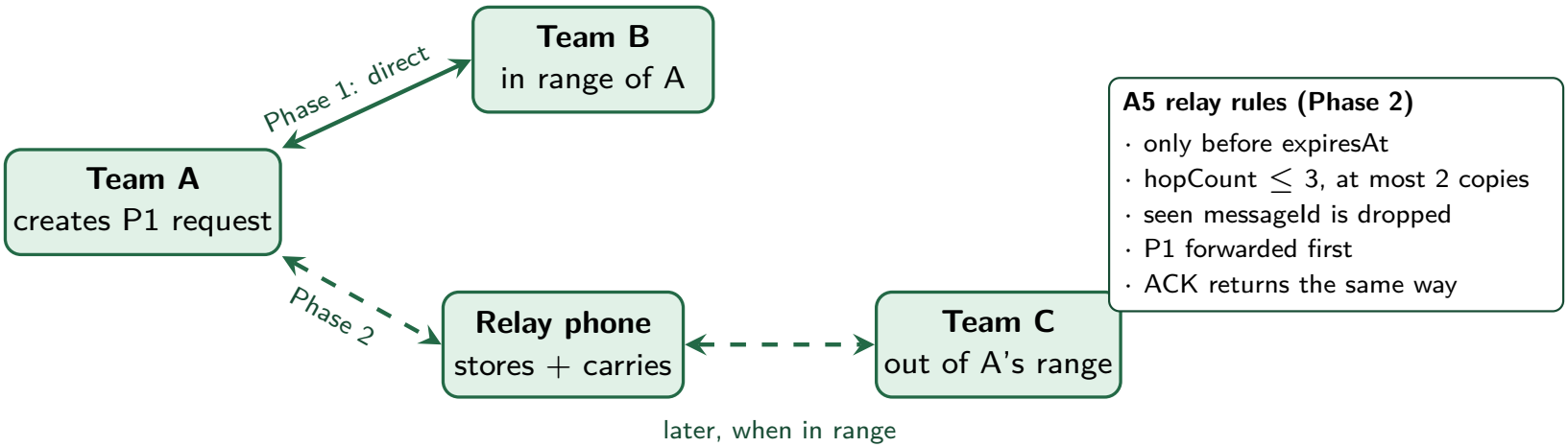
Safeguards (planned): authorised devices only, minimal patient fields, encrypted storage and channel, audit metadata.

If the link fails: the record stays queued and is retried; the crew sees it is not yet delivered.

Outcome: the desk receives structured details with a confirmed state.

Limits: no hospital-system integration; an ACK means saved, not read.

Use Case 3: Rescue in Damaged Infrastructure



Phase 1 without a relay: Team C's request stays QUEUED at Team A until C is in range, or it EXPIRES.

Situation: after a building collapse, teams work in separate zones; concrete and distance block direct links.

Actors: field teams A, B, C; incident coordinator.

Constraint: not every team can reach every other team directly.

Algorithms: A1–A4 in Phase 1; **A5 only in Phase 2**, a design to be built and tested, not something Wi-Fi Direct provides.

If the link fails: the message is kept and the relay copy is also bounded, so nothing floods the area.

When it returns: the destination validates, saves once and ACKs; duplicates from two paths are dropped.

Outcome: more requests reach the right team, each with a known state.

Limits: with no relay in range, delivery is not guaranteed; stale requests expire and are flagged.

Impacts and Benefits

- **Benefit to the target users:** requests stay available offline, urgent cases go first, and senders can see whether each request was saved by the receiver.
- **Social and economic impact:** supports disaster preparedness and coordination using phones teams already own; may reduce duplicate or missed requests.
- **Expected measurable outcome** – *targets, not results; no tests run yet:*

Metric	Method (dummy data, 4 phones, no internet)	Target
Delivery success	sender holds ACK within 5 min of a link being up; 500 messages per condition	$\geq$ 95%
ACK latency	send to ACK, at 5 / 20 / 40 m after a separate range test	median and p95 reported
Duplicate incidents	every message re-sent 3 times; count extra incident rows	0
Altered frames	100 deliberately corrupted or unsigned frames	100% rejected + logged
Lost ACK and restart	drop ACKs; kill app mid-queue; 10-min out-of-range gap	no loss; re-ACK; resume
Expiry	messages with short expiresAt	stop at expiry, flagged

Delivered = the receiver committed the message and the sender recorded its ACK. Relay (A5) is tested only once built.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 3	Target 3.d	Good Health and Well-Being	3.d: capacity for early warning, risk reduction and management of health risks. The system aims to keep emergency medical information flowing during outages.
SDG 11	Target 11.5	Sustainable Cities and Communities	11.5: reduce deaths and people affected by disasters. The system aims to help responders coordinate rescue requests when networks fail.

Contribution is indirect and depends on successful implementation, validation and adoption.

Individual Member Contribution

Member	Contribution or responsibility
Palella Bhanu Prasad	Project coordination, system architecture, A1 prioritisation and final integration
Gullanki Bhagya Lakshmi	Wi-Fi Direct link layer, offline networking workflow and Phase 2 relay design (A5)
S V Pavan Kalyan	Request data model, Room database, location capture and A2 store-and-forward
V Manasvith Reddy	ACK and retry logic (A3), deduplication and integrity checks (A4), test plan and evaluation

Yukti ID

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